

Hopper the Grasshopper · 3D edition

Design and production plan

A 3D action game about an enormous insect robot, the small boy on his back, and the freedom of a leap that clears a tower. The player crosses vast painted districts by hopping from structure to structure, fights shadow creatures on rooftops and in the open air, and keeps a far landmark on the horizon that grows closer with every chapter. Falling never hurts; only shadows do.



The delivered Hopper and rider model standing in a stand-in Sunseed Fields, with the Crownline skyline on the horizon.

This document reinterprets the shipped 2D game as a 3D game. It keeps the title screen, the three music recordings, the three missions and nine regions with their characters, the eighteen shadow species, the three commanders, and the 1970s cel-and-gouache personality. It changes everything about how the game plays: the 2D rules are not ported. Only ideas that earn a place in three dimensions return, and most of the movement and combat is new.

Three things are already real. The animated Hopper, rider and combined models are delivered in `hopper/models/`. Every other model, texture and sky the design needs has a procedural stand-in in `hopper/3d/standins/`, rendered in the figures of this document, so building can start today and art can arrive in any order. The full asset requests, each with the rigging it needs, are in `model-requests.md` and `image-requests.md`, and `standin-manifest.json` ties every request to its placeholder and its final file.

All numbers are starting points for the first playable, chosen so that the feel described here is reachable on the first day of tuning. The next step is a review of the play direction, the control mapping and the

production sequence, then milestone one.

The experience

The first thing the player does is press A and hold it. Hopper's hind legs compress, snap open, and the ground drops away for four and a half seconds. At the top of the arc the fields are a quilt, the farmhouse is a toy, and on the horizon, past the terraces, the ivory towers of Crownline City are visible. Keep holding, and the wings open: Hopper glides, losing height slowly, steering with the stick, for as long as it takes to reach the next terrace, or the silo cap, or the roof of the seed vessel where three hounds are waiting. Land badly and there is no penalty but the climb back. Land on a hound and it dies, and Hopper bounces off it higher than the jump that got him there.

That is the loop: read the space, choose a line, leap, fight whatever is on the way or in the way, land, and leap again. It is played at Hopper's scale, fourteen metres to the head, in districts two and a half kilometres across, with the region's exit standing on the horizon from the moment the region starts.

Design pillars

- **The leap is the game.** Height, hang time, gliding and air steering are generous. Every roof, mast, crane and reef is reachable, usually by more than one route. The joy is choosing lines.
- **Falling is free. Only shadows hurt.** There are no pits and no fall damage. Every district has a floor, and every low place has a way back up within about twenty seconds. Armour is lost only to enemy attacks. Hazards push, lift and fling; they never damage.
- **Fight on the ground and in the air.** Flyers are steps: stomping one bounces Hopper upward, so an air fight is a chain of bounces, kicks and laser bursts that can keep him aloft. On the ground the fight is about position, cores and timing, and the same buttons.
- **A far thing that gets closer.** Each region shows its exit landmark from its entrance. Chapters bring it nearer; the last chapter reaches it. The player always knows where the story is going because it is on the horizon.
- **The same face.** The same title screen, logo, poster and music; the same Hopper, rider, shadows and commanders; the same painted, hand-inked look, now built from cel-shaded models under painted skies.

Rhythm and length

Mission	Regions	Play target
1 Earthbound Thunder	Sunseed Fields · Crownline City · Thunderhead Range	28–34 min + Night Rook 4–5 min
2 The Iron Migration	Cinder Foundries · Tempest Docks · Skyhook Works	30–36 min + Smelter Leviathan 4–5 min
3 Beyond the Black Sun	Vermilion Basin · Cobalt Drift · Violet Inversion	32–38 min + Eclipse Regent 5–6 min

A region is about ten minutes on the main line and longer for anyone who chases signals. Each region is five named chapters, keeping the 2D chapter names, and every chapter is a leg between two landmarks with one dominant beat: learn, run, fight, climb, drop, hazard, vista, rest or finish. The whole campaign is around one hour forty-five minutes, with shorter repeat clears.

What stays and what changes

Kept from the 2D game	Reinterpreted for 3D
Title screen: poster, logo, Press Start, play-select with the three episodes	Same components; a 3D rendered poster is optional later
Music: the title theme, Grass March through episodes one and two, Dark Moon through episode three	Same routing; effects gain 3D position
Three missions, nine regions, forty-five chapter names, three commanders	Regions become open districts with a landmark spine instead of a corridor
Eighteen shadow species with their silhouettes, tells and cores	Behaviours rebuilt around rooftops, air lanes and stomp bounces
Hopper, the rider, the eye lasers, the hind-leg kick, the stomp	Now a full 3D move set: glide, dive, wall kick, mantle, crouch charge, lock-on
Armour pips, checkpoints, signals, caged signals, capsules, spring pads	Spring pads, thermals and wind lanes carry Hopper up rather than across
Heavy, low and inverted gravity in the third mission	Gravity volumes in an open space; inversion galleries on arch ceilings
1970s cel character treatment and gouache landscapes	Cel-shaded models with ink outlines under painted equirect skies
Xbox-first controls, menus, pause, settings, accessibility rules	Camera controls and 3D readability aids added

Deliberately not carried over: the directional two-way defence of the 2D game (rear kick only, front guard only) and the held guard with it, the one-way shelves, ceiling runs as a required mechanic, the fixed side view and its camera rules, and death by falling. Those were answers to a side-scroller's questions. The 3D game asks different ones.

What the player is doing, minute to minute

1. **Reading the district.** Where is the landmark, which structures lead toward it, where are the shadows perched, where is the way back up if this jump misses.
2. **Choosing a line.** A tap-hop across roof decks, a full leap to a tower ledge, a wall-kick ladder up a corner, a glide off a chimney onto a train, a spring pad into a thermal.
3. **Fighting through it.** Shooting the sniper on the far roof while gliding in; kicking the pouncer that lands beside him; stomping a ray to bounce over the gap; diving onto a nest.
4. **Landing and looking up.** A checkpoint totem lights, the landmark is nearer, the next leg is visible.

Hopper in three dimensions



The delivered hopper-rider.glb in its Idle clip, rendered at 1:1 in a stand-in field.

The delivered models set the scale of everything. Hopper is 14 m from his feet to the camera socket on his head, 29 m long from the antennae to the hind feet and 23 m wide across the leg spread. The rider is 5.6 m tall, an artistic fit to the couch rather than a child's height, and is parented under the seat socket in the combined file. Call Hopper's height **H**; the movement numbers below are in metres and in H.

What the models give the game

- **Rig:** a 51-joint insect skeleton with six legs, two wings and three antenna segments per side, and a 41-joint humanoid rider. Both face +Z, +Y up, metres.
- **Clips:** twenty Hopper actions and five rider reactions, listed in `models/manifest.json` with their event timings. Walk and Run are in place; the jump family is physics-driven.
- **Sockets:** feet, laser origins, camera, centre of mass, rear-kick hitboxes, seat, grips and footrests; on the rider, hands, feet, chest, head, camera and mount.

Clip mapping

Game state	Clip	Notes
Standing, looking	Idle	Rider Riding_Idle layered

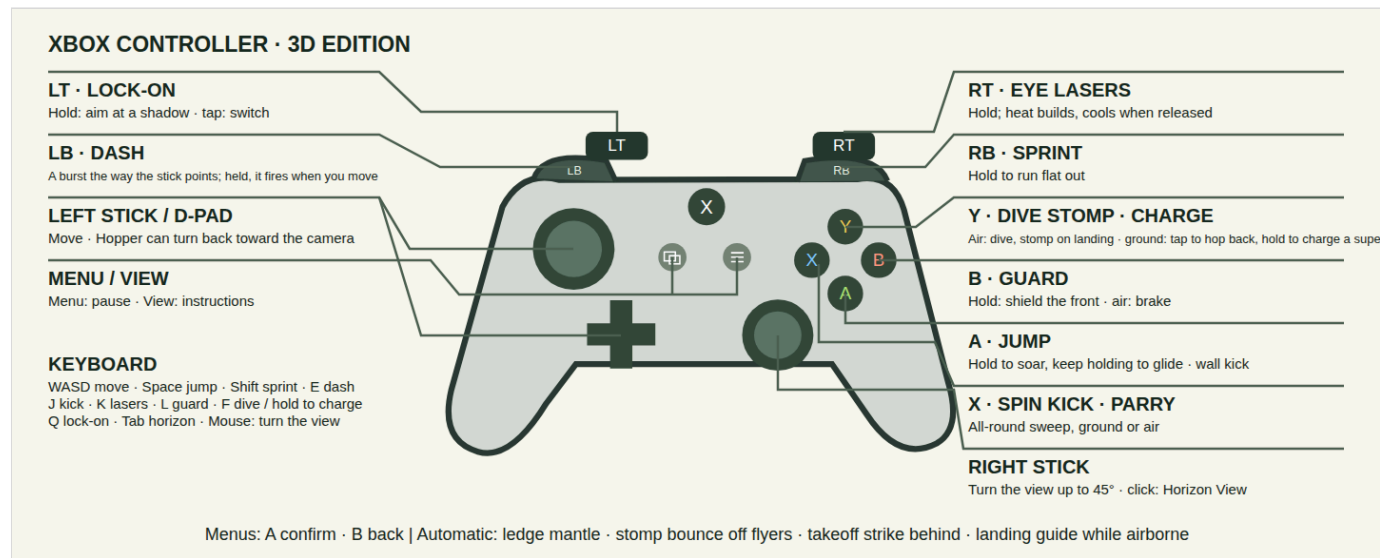
Game state	Clip	Notes
Moving	Walk → Run by speed	Blend by stick deflection
Crouch charge	Crouch → Crouch_Hold	Femurs compress with charge
Takeoff	Jump_Start	Rear attack window 0.22–0.6
Airborne	Jump_Loop	Physics owns the arc
Glide	Glide_Loop (new)	Wings open over 0.25 s
Dive	Dive_Loop (new)	Legs tucked, head down
Landing	Land / Stomp_Land (new)	Ground contact event 0.16
Kick	Spin_Kick / Air_Kick (new)	Facing preserved
Lasers	Fire_Start → Fire_Loop → Fire_End	Laser sockets aim the bolts
Hit, defeat, win	Hit_Reaction, Defeat, Victory	Rider Hit_Reaction, Cheer

The new clips are request M-001; they add to the existing rig without changing the mesh. Locomotion, though, is not a clip any more and will not become one: walking, galloping, climbing and flying are driven procedurally over the delivered skeleton, because feet that are placed on the world can hold a slope, a stair and a wall that no authored cycle can. Clips are kept for what the whole body does at once — the kick, a blow taken, the lasers, the crouch — and the procedural pass fades out under them while they play. It runs after the animation mixer, never before, since the mixer writes the whole skeleton every frame; the wings are beaten there too, which is why a hover drums steadily rather than fluttering.

The rider

The boy is company and a scale reference, never a health bar. He leans into leaps, braces for stomps, streams his scarf in a glide, points at the landmark during Horizon View and cheers when a knot is cleared. He has no separate hit points and cannot be knocked off.

Controls



Xbox controller diagram for the 3D edition.

Xbox	Action	Keyboard and mouse
Left stick / D-pad	Move, camera-relative; full deflection runs	WASD
Right stick	Camera; click to reset behind Hopper	Mouse
A	Spring: a tap under 0.3 s is a hop taken without breaking stride, a longer hold stops Hopper and winds the jump and the stick aims it; hold in the air to hover on beating wings, keep holding to glide; at a wall, wall kick	Space
X	Spin kick, all-round, ground or air; parries a shot from any side	J
B	Spin kick, the same sweep on the other thumb	L
Y	In the air: dive, stomp on landing. On the ground: hop back	F
RT	Eye lasers (hold); heat builds and cools	K or left mouse
LT	Lock-on (hold); tap to switch target	Q
RB	Sprint (hold)	Shift
LB	Dash, the way the stick points, on the ground or in the air	E
Right-stick click	Horizon View (hold): frame the far landmark and the next checkpoint	Tab
Menu / View	Pause / instructions	Escape / I

Automatic, with no button: ledge mantle when the front feet reach a lip, stomp bounce when landing on a flyer or an exposed back, the takeoff strike behind Hopper in the first 0.16 s of any jump, and the landing guide projected below him whenever he is airborne. Menus keep the collection's conventions: A confirms, B backs out, any fresh button starts from the idle title screen, the left stick enters utility navigation.

The mapping keeps the 2D game's muscle memory where it still means the same thing (A jump and hover, X and B kick, RT lasers, Menu, View) and gives the unused inputs to the new 3D verbs. Nothing essential is on a combination.

Movement

The simulation runs at a fixed 120 Hz, as the 2D engine does, with animation timed independently. Gravity is stylised at 120 m/s² in Earth regions, 1.25× that on the way down, far heavier than the real thing so that a jump is a launch and a landing has weight: the same apexes as before in about half the airtime.

Parameter	Value	Result
Run speed	66 m/s (4.7 H/s), 106 sprinting	Crosses a 100 m roof in a second and a half
Acceleration, stop, turn	0.2 s · 0.15 s · the camera's forward	Hopper faces the route; backpedal at 55 %, strafe at 85 %
Tap jump	23 m apex (1.6 H) in 0.6 s, 1.15 s airtime; 108 m running, 162 sprinting; a 0.2 s hold adds a few metres	A quick launch between roof decks
Hover	A held in the air: 1.8 s of altitude held on beating wings, refilled on landing	Cross a gap, line up a landing, fight above the ground
Takeoff lunge	+24 m/s along the run (or the stick)	The leap goes forward, not merely up
Glide	Sink 7 m/s, forward 80 m/s, once the hover is spent	Height becomes distance at eleven to one
Crouch charge	0.8 s to full; super leap to 140 m (10 H), 6.1 s	Reaches a tower roof from the street
Spring pad	Launch to 168 m (12 H); hold A to float	The tallest single climb
Thermal	Lift 25 m/s inside the column while gliding	Chimneys, furnace doors, vents
Wind lane	15 m/s lateral push, marked by streaks	Bends a glide, never past its landing
Stomp bounce	56 m (4 H); 84 m with A held	Flyers are steps
Dive	2.5× gravity, terminal 190 m/s; 100 m in 1 s	Fast return to the ground, stomp on landing
Wall kick	62 m/s up (16 m), 26 m/s away, unlimited chain	Tower corners are ladders
Climb	Push into a face and he takes hold: 27 m/s up, 19 m/s along it, no fall while he holds	Any wall is a way up; A kicks off it
Ledge mantle	Reach 6 m over the lip, 0.7 s	Forgives a short leap, and tops out a climb
Air control	Half of ground acceleration; steering turns the arc, never slows it	Momentum is kept; pushing back still brakes
Coyote time, input buffer	0.12 s · 0.14 s	Forgiving edges and landings

How he moves: three gaits

A grasshopper the size of a horse should move like one when it is crossing open country and like an insect when it is picking its way up something, so there are three gaits and the ground chooses between them.

- **Gallop.** Open ground at a run: four-beat, hind pair first, then the middle and front pairs, with a moment in every stride when nothing is touching the ground. The legs gather in under the body where a horse's are, the body runs low with the knees kept bent, and it bounds — the barrel rising and falling once a stride, the nose lifting as the hind legs drive and dropping as the front pair take the landing. It needs to be a way of going forward: crossing the ground sideways or backing off is the walk, however fast.
- **Walk.** Anything slower, and anything on ground steep enough to pick over: the insect's alternating tripod, three feet down at all times, the wide stance, the high step.
- **Climb.** Ground past about 46°, and any wall he is holding: the same tripod, slower, splayed wider, hugging the face.

They cross-fade by speed and by the angle of the ground, so there is no moment where he changes gait; the stride clock is shared, so the feet never jump when one gives way to another.

The feet are put on the world, not on the body. A planted foot stays exactly where it was put while the body travels over it, and a swinging foot arcs to where the body will be; the renderer solves two-bone inverse kinematics up each leg to reach them. Nothing is authored per surface, so a stair, a boulder, the lip of a roof and the face of a silo all come out of the same solve, and the body reads its own pitch and roll back off the ground under its six feet rather than being told. In the air the legs hold what the leap left them with: the hind pair swept back and down behind him where they finished the shove, the front and middle pairs drawn up, and everything reaching forward again as the ground comes up.

The jump family

A tap is a leap, not a hop: takeoff adds a forward lunge along the way Hopper is already running, and the air steering redirects that speed without bleeding it, so a jump goes four to seven times further than it goes up. A short hold after takeoff adds a few metres, no more. Holding A in the air is a **hover**: the wings beat hard, the fall stops, and Hopper holds his height for just under two seconds, long enough to cross a gap, pick a landing or trade shots with a flyer. There is no second jump and no boost; the hover is a pause in the air, not a climb. When the hover is spent and A is still held, the wings lock open into a glide that lasts until Hopper lands, presses Y to dive, or releases A to drop. Releasing and pressing A again while falling hovers again if any fuel is left; the fuel refills on the ground. Steering in the air is strong enough to correct a line, never so strong that arcs stop reading.

Hopper always faces forward: along the trail, or wherever the camera's manual turn points, or at a locked target. Pulling the stick back is a backpedal at half speed, sideways is a strafe, and the dash, the hop back and the wall kick are manoeuvres that keep the facing. Turning round is the camera's job, never the stick's, so the rider is always looking where the danger is.

As built: the jump is a spring. A held on the ground roots Hopper where he stands and winds it up; letting go launches it, and the fall pulls harder (1.4×) so the arc is quick up and sharp down. A quick tap barely winds and is a small hop taken without a pause; a second and a half is the full charge. The wind sets one launch speed and the stick sets the angle it leaves at: fully forward is 15° above the ground, a flat dart

that reads as a lunge rather than a jump; neutral is 45°; fully back is straight up, and a half push lands halfway between. Pulling back never throws him backwards — the backward part of the stick only stands the launch up, and whatever forward or sideways part is left gives the direction. A tap cannot reach the flattest angle (25° at no wind), so the lunge is earned, and the spring never leaves slower along its own line than he was already travelling, so a tap at a sprint keeps the sprint. At 68 m/s that gives a 4 m apex over 40 m for a forward tap, 15 m over 192 m at three quarters, 16 m over 296 m at full wind and 325 m/s, 120 m over 561 m held neutral, and a 240 m climb pulled fully back, which is the highest jump in the game. The crouch reads the whole thing in advance: he sinks deeper as the spring winds and tips back toward wherever it points, and at full wind he throbs against the stop inside a ring of light while the pad knocks once. Holding A in the air is still the hover and then the glide. Every number that shapes the jump lives in `hopper/game/src/game3d/jumptuning.ts`, which the controller spreads into its movement constants, so the feel is retuned in one file rather than in the simulation.

The crouch charge is the deliberate big jump. Holding RB compresses the hind legs over 0.8 s; releasing launches straight up by charge, and the stick during the charge sets the direction. It is slow to start and enormous, so it is the way onto the tall things and rarely a combat move.

Wall kicks chain without limit, because the world is full of corners and the ladder up a tower is a rhythm the player should be allowed to enjoy. Each kick preserves facing and commits steering for 0.16 s.

A wall can also simply be climbed. Pushing into a face takes hold of it, from a standing start as much as out of the air: gravity stops, the stick's push into the wall is the climb and its slide along the wall the traverse, and letting go of the stick leaves him hanging and sliding slowly rather than falling. A is the wall kick off it, pulling hard away from it or pressing Y lets go, and a lip within mantle reach is hauled over as it always was. Hopper lies against the face with his back to the camera while he is on it.

The trail

Every district is a trail: a smooth route from the start totem through each checkpoint to the exit, authored as waypoints so that it winds between the structures, climbs the terraces and plateaus, drops through the valleys and keeps going forward. The ground is painted with a worn path along it and dressed with a ribbon the width of Hopper's stance, stones along both edges, a lit waymarker every 55 m, and tall things standing 20–95 m back from the edges (trees and haystacks, pylons and lamps, rock spires and cairns) so the scale of a 14 m grasshopper and the distance still to go can be read at a glance. Hopper may leave the trail whenever he likes; it is the direction the camera faces and the way home, not a corridor.

Districts hand over across a **threshold**: a lit arch where the trail ends, with the next district visible through it on the horizon in its own colours. The district ahead is not a painting: its terrain and its structures are built at a lower level of detail and placed so that its start sits exactly on the current exit, so the city grows on the horizon from the first chapter, fills the view as the last stronghold falls, and is the same city that stands there after the crossing. Only the final district of a mission shows a silhouette landmark instead. Crossing it dissolves the last frame of the old district into the new one over about a second, and Hopper arrives with the heading, the speed and the airborne state he crossed on, measured from the trail so that forward stays forward. A region change should read as walking on, not as being put somewhere else.

Either side of it the country is filled in: clumps of the region's tall things out to 430 m, and real structures — farmhouses, towers, crag columns — standing well back from the path with their own colliders, so what looks climbable is climbable. Every span lands on something: a bridge or an elevated rail carries an

abutment under each end of its deck and one more span beyond it, because a crossing that starts and stops in mid-air reads as scaffolding rather than a world.

Falling is free

There is no death fall. The lowest surface in every district is ground, water, slag or dust, and each of those returns Hopper to play: water and dust are soft landings that push him back toward shore, slag is a hot updraft that lifts him out. A missed leap costs the climb back, and the design rule is that from any point in a district a spring pad, thermal, stair of structures or wall-kick corner leads back to the main line within twenty seconds. Signals are hidden at the bottom of things on purpose, so that falling is sometimes an invitation.

Knockback from an enemy hit can throw Hopper off a roof. That is the real cost of a mistake, and it is never fatal by itself.

Camera and readability

The camera follows from 54 m behind and above the centre-of-mass socket with a 66° vertical field of view, so Hopper fills about a fifth of the frame and the rider stays visible on his back. It faces the way onward: along the district's trail, the route's own tangent 80 m ahead of Hopper's place on it, never toward a point, so it pans only as the trail bends (never faster than 0.9 rad/s) and never faces back toward the start. Hopper is free to turn round and run toward the camera; the view holds its direction. The right stick turns the view up to 144° either way, and forward is the default rather than a demand: a turned view stays where the player left it, and only after two seconds without a touch on the stick does it begin to drift back, coming up to speed over another two seconds and never turning faster than about $26^\circ/\text{s}$, so the player never feels the camera fighting them. Clicking the stick brings it round in under a second, as a pan rather than a cut. Camera collision is a spring against structure colliders, never a cut: a blocked view pulls the eye in quickly and lets it back out slowly, and the ground lifts the eye rather than pulling it in.

The camera takes its own smooth path and Hopper moves within the frame. Its look point leads him by his velocity as the camera sees it, smoothed over a few tenths of a second, so a landing, a dash or a wall kick changes the framing at the camera's pace rather than the body's; and it follows him more slowly in height than along the ground, slower still while he is airborne, so a hop and its landing ride on one arc rather than bobbing the whole picture. Hopper drifting to one side of the frame is fine; the camera jumping is not. Only a real change of level, a roof gained or a long fall, moves the picture up or down, and a wide gap closes faster.

- **High leaps** pull the camera back and tilt it down as height above the last surface grows, up to 55 m behind at a super-leap apex, and ease back in as Hopper descends toward a landing. The landing is always framed before it is reached.
- **Gliding** lowers the camera to just above wing height and widens the field of view by 8° , so distance and speed read.
- **Diving** looks down the dive with Hopper high in the frame, so the landing and whatever is on it are visible.
- **A commander in the air** lifts the camera's look toward it and pulls the view back while it is awake, so a boss that fights from 90 m up is framed with Hopper rather than off the top of the screen. Its lasers and lock-on treat it as an ordinary target.
- **Aiming (LT held)** is a camera mode of distance and zoom, never of direction. The camera comes in from 54 m to 46, steps over Hopper's right shoulder so his flank is clear of the middle of the screen, levels the picture off toward the horizon and narrows the field of view from 66° to 46° ; the right stick slows to a little over half speed so a shadow three hundred metres out can be held. The view points where the player points it, going in, coming out and throughout: taking a target, cycling, killing one or releasing the trigger never turns it a degree, and the trail bending underneath does not drag the shot off either. The crosshair is the middle of that picture, drawn in the world so it reads at any range. A shadow the aim comes near takes hold of it and it slides the short way onto that shadow, keeps it while the aim stays roughly on it, and lets go to the centre when the shadow falls or the aim leaves. Hopper faces where the camera faces, so the shot follows the crosshair and the left stick strafes. Tap LT to cycle

between the shadows near the crosshair. Lock-on is soft: lasers auto-aim in a cone whether or not a shadow is held.

- **A climb** turns the view round to stand off the wall and watch his back go up it, and holds its pitch inside a narrow band: climbing straight up must never point the camera at the sky, so the eye stays a little above the look point, level with the face, and the picture never cranes after him.
- **Horizon View (LB held)** swings the camera up and out to frame the region's exit landmark, the next checkpoint totem and any signals within 300 m, with distance labels. The rider points. Releasing keeps the direction it was left at, as a manual turn, and pans it back along the trail within a second: the return is a turn, never a cut. It is the "which way" button and also the "look at that" button.

Reading the ground from the air

Three aids make 3D landings as certain as 2D ones. A **landing guide** ring is projected onto the surface directly below Hopper whenever he is airborne, scaling with height, with a second predicted ring where his current velocity will land him. **Height ticks** on the HUD edge show height above the surface below. Structures have a **flat-top language**: every landing surface is a flat plane in a lighter tone with a dark lip, and slopes above 35° are visibly rock, glass or plating that cannot be stood on. The stand-ins already follow this, and every structure model request lists its landings.

Reduced motion halves camera pull-back and disables the glide field-of-view change. Camera sensitivity, inversion and field of view are settings.

Combat

Shadows are built to Hopper's scale or beyond: a hound stands eye to eye with a 14 m grasshopper, a tortoise is a moving hill, a condor's span is three of him. The enemies of a giant have to be worth fighting, and a fight reads from across the district. Only shadows can hurt Hopper. He has six armour pips; ordinary hits cost one, heavy and boss hits cost two, and 1.1 s of soft-outlined protection follows a hit. Capsules restore two, checkpoints and arena clears refill all six. Reaching zero plays the defeat and restarts at the last totem within three seconds, with the chapter's shadows reset and its cleared knots kept.

Attack	How	Damage and effect
Eye lasers	Hold RT; auto-aim within a 30° cone, 250 m	1 per pulse, 8 pulses/s; heat locks after 3.5 s, cools in 2 s
Spin kick	X or B, ground or air; 16 m past the body and 6 m above and below it, all round, drawn as a bright sweeping arc	4; parries a projectile or blow from any side through most of the swing
Dive stomp	Y in the air, then landing	5 in a 12 m shockwave; knocks ground shadows into the air
Stomp bounce	Land on a flyer or an exposed back	5; rebound to 56 m, 84 m with A held
Takeoff strike	Automatic, first 0.16 s of a jump, 10 m behind	3 and a stagger
Parry and reflect	Kick as a shot arrives	Shot returns to its shooter for 6; opens cages; staggers the shooter
Hop back	Y on the ground	0.2 s of invulnerability and a 12 m step away

The kick is all-round because in 3D the question is timing, not direction, and it is the whole defence: there is no held guard in this edition, as there is none in the 2D game any more. A dome that has to be raised and aimed asks the player to think about facing at the moment they most need to think about timing, and it made the kick the lesser answer to a volley. Instead the kick's parry window is generous and its reach is long, the sweep is drawn so its cover is legible, and it sits on both X and B so the nearer thumb can meet a shot.

Fighting in the air

The fight comes up at Hopper, not down on him: once he is near, window rays, rift condors and chain mantas drop below him (never lower than a body above the ground, never above their own perch) and their dives rise to meet him, while the rooted shooters lob and beam upward from where they stand. The super jump and the way down — stomps, kicks, dives — are the answer; aiming upward is never the game.

Every flyer is a step. Stomping one damages it and bounces Hopper upward; holding A on the bounce goes higher; releasing takes a low, controllable rebound. Kicks work in the air and preserve the arc. Lasers fire while gliding, so a glide toward a roof can clear the sniper on it before arrival. Pushing the stick back against the flight brakes it, which is a dodge as much as a steering correction. A chain of bounce, kick, bounce can keep Hopper aloft through a whole flock, and the flocks over the ravines and reefs are built to invite it.

Fighting on the ground

Ground shadows attack with clear tells 0.5–0.9 s long, and the ground game is about cores: a hound's ribs open when it lands from a pounce, a tortoise shows its belly in a lunge, a crab shows its rear vent after a slam. Lock-on, strafing and the hop back are for getting to the core; the dive stomp is for opening a group at once, because it knocks ground shadows into the air where they are helpless for a second and can be kicked or shot.

Pacing: interludes and strongholds

The HUD names what is being aimed at and shows its health: the commander gets the full bar, phase and tell; anything else locked or under fire gets a small name-and-health readout, so a player can always tell whether the shots are landing.

A district alternates two kinds of ground. An **interlude** is three or four hundred metres of clear trail: scenic props left and right (windbreaks, farmhouses, silos, crag columns, roof decks), a totem at its start, one or two jump elements on the path itself (a terrace to climb, a shelf sequence, a spring pad over a gap) and at most one patrolling pair of shadows, so the road is alive but not a fight. The next battle is always in view: a **stronghold**, a cluster of tall things (a granary of silos, an ivory tower pair, a ring of crag columns around the mast) that the interlude walks toward and that grows on the horizon.

A stronghold's **host** is in plain view long before the fight. Every member waits on the nearest structure top near its spawn (the game finds the perch: the closest top within 70 m standing at least 10 m up, and most of a host is 20 m up or more), crouched, turning to face Hopper from 700 m out, its eyes brightening over the last 500. The player sees the silos with hounds on their tops and the tower crowns with rays folded on them, and knows exactly where the fight will come from. Coming within the stronghold's radius brings the host down over five or six seconds in an authored order: hounds leap off their perches at Hopper, rays and condors launch from the crowns for their stations, spitters and leeches on the faces wake in place, one ambusher waits until Hopper has passed and leaps at his back, and a second surge comes down when the first is down. A perched shadow can be shot or stomped before the stronghold wakes, and doing so brings that one down at once. The HUD counts the host. When the last of it falls the region is **freed**: a banner, two hearts back, a thousand points. Nothing bars the trail: a stronghold is a dangerous place on the way, a dense cluster of buildings or a pass with things waiting in it, never a door, and the threshold to the next district is open whether or not the hosts were cleared. Only the mission's commander holds a way shut: Thunderhead's last stronghold is the Night Rook's arena, and the episode ends when it falls.

Nothing arrives from nowhere: every host member was visible on its perch before it moved, and every arrival has its tell, the crouch on the perch, the launch off the crown, the stare that brightened as Hopper came closer.

Enemy health and scaling

Light shadows have 4 health, ranged 6, armoured 10–12, in units of laser pulses. Difficulty grows through waves and the mix of species, never through health padding. Standard and relaxed modes exist as in the 2D game; relaxed lengthens tells and halves incoming damage.

The eighteen shadows in 3D



Stand-in silhouettes for all eighteen species on a contact sheet; the final models are requests M-003 to M-020.

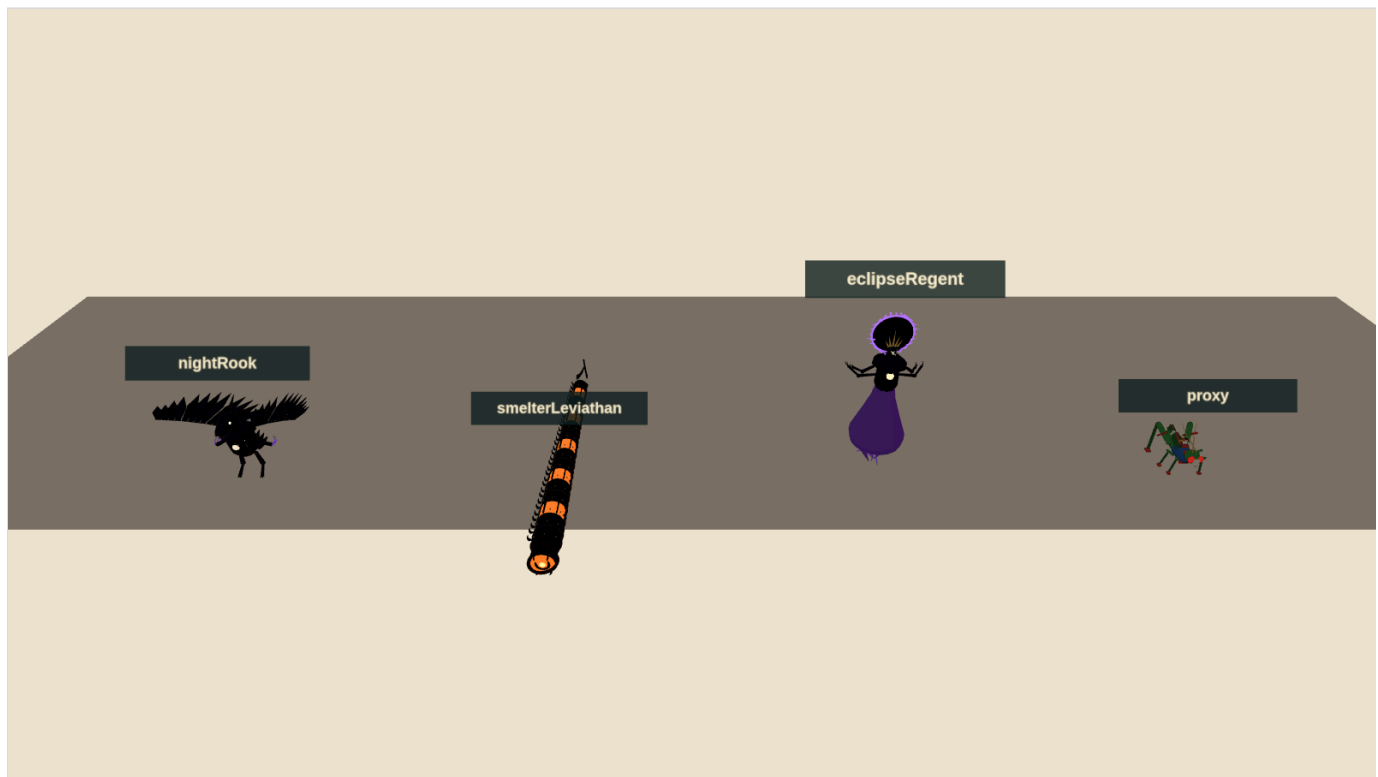
The shadows are drawn as dark spots on the picture: an optional pass (the "Shadow halo" setting, on by default) renders their silhouettes to a mask against the frame's own depth, blurs it at a quarter of the resolution and lays it back over the frame as a feathered violet darkness that hugs each silhouette and bleeds a little into what stands around it. Nothing inside the silhouette changes but a touch of darkening, so cores, eyes and tells still read; a shadow behind a wall leaves no spot.

Species	Region	Role in 3D	Tell and counter
Shade Hound	Fields	Pouncer that follows Hopper's line across terraces and roofs	Crouch, flash, pounce; ribs open on landing: kick, or stomp the back
Seed Spitter	Fields	Rooted sniper on terraces and roof gardens	Throat inflates, three arcing seeds; lasers between volleys, or glide in and kick
Window Ray	City	First flyer; hangs between towers	Pause, fold, straight dive; jump over it and stomp the back
Spire Leech	City	Wall-clinger that beams across gaps	Eye charge, horizontal beam; flank from another face, or reflect a shot
Crag Tortoise	Mountains	Armoured shelf-holder	Rears and lunges uphill, belly shown; never stomp the spikes

Species	Region	Role in 3D	Tell and counter
Rift Condor	Mountains	Thermal-rider that duels in the air	Feather corridor, swoop; bounce off its sternum, or kick as it climbs out
Furnace Hound	Foundry	Straight-line charger along belts and decks	Vents, charges; jump it and strike behind, or stomp mid-skid
Slag Caster	Foundry	Lobber whose slag becomes stepping stones	Scoop, lob; core opens after each throw
Chain Manta	Docks	Tether-puller that drags platforms	Tether drops, load slides; shoot the tether node, bounce off its back
Ballast Crab	Docks	Deck-slammer with a shockwave	Rears, slams; jump the wave, hit the rear core
Coil Wraith	Launchworks	Lane gate between two nodes	Nodes brighten, beam holds; shoot a node out
Turbine Wasp	Launchworks	Dasher that becomes a platform	Intake contracts, dash; kick to stall the fans, then stomp
Basalt Burrower	Basin	Erupts under the ground you land on	Cracks trace it; jump the eruption, stomp the soft back
Thorn Choir	Basin	Fan-volley chorus on ivory arches	Heads open in sequence; kick one to stagger all
Veil Medusa	Drift	The springiest stomp in the game	Bell flashes, radial pulse; lasers pierce the bell; stomp it for a high rebound
Phase Skate	Drift	Blink-dasher	Fades, silhouette, solid, dash; free stomp for 0.4 s after it solidifies
Mirror Stalker	Inversion	Floor and ceiling lunger	Mask tilts, mirrored lunge; its stance shows its surface
Gravity Cantor	Inversion	Local gravity flipper	Prongs count in, volume flips; shoot the core or fly around

Each species keeps its 2D silhouette, its core and its tell, so a returning player recognises everything. What changes is the space: snipers hold roofs, chasers use Hopper's own routes, flyers own lanes between structures and are also his stairs.

The shadow commanders



Stand-in bosses beside the Hopper proxy for scale: Night Rook, Smelter Leviathan, Eclipse Regent.

Night Rook at the transmitter

The mission-one boss fights around the summit mast, on three shelves at different heights and in the air between them. At full health it perches on the mast, marks a dive corridor with a feather trail and sweeps past; the player leaps over the sweep and bounces off its back, or shoots its sternum as it climbs out. It fans blade feathers from the mast top, leaving a gap that has to be glided through. At two-thirds it lands and channels through its wing joints; kicks break the guard and open the core. At one-third the fight goes airborne: the Rook stays aloft and the player fights it entirely by bouncing between condors it summons (capped at two) and the Rook's own back. Its defeat dissolves it around the mast, which lights.

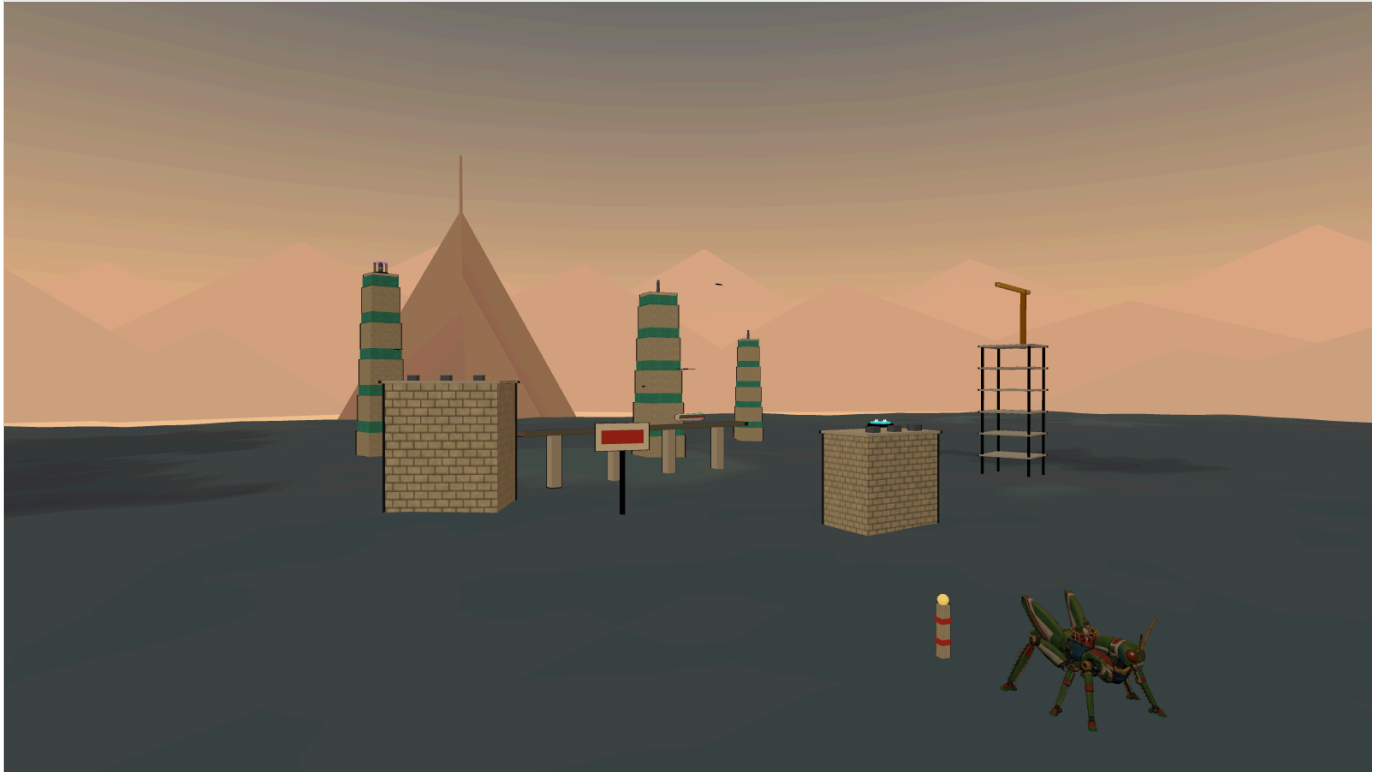
Smelter Leviathan on the gantry

The mission-two boss is coiled around the gantry elevator tower with the launch ring above. The fight climbs. On the pad the tail sweeps the deck; jump it and kick the exposed joint. Its furnace breath fills one marked side of the tower; wall-kick up the other side and fire into the cooling maw. Higher, it uncoils across the ring pads and exposes cores along its length in sequence; stomping a cracked segment opens the inner core for lasers. The elevator cage is the safe platform between phases, and falling off the tower lands on the pad with a thermal back up. Its last core stops the engine and opens the star gate.

Eclipse Regent at the cathedral

The final fight is on the eclipse dais before the cathedral facade, in three gravity phases announced by seams and a 1.5 s count-in. Heavy gravity first: crawling shadow rings cross the dais; leap them and kick the lowered hands to open the heart. Low gravity second: marked lances fall between the six mirrored shelves; summoned medusae (capped at two) are the bounces up to heart height. Alternating gravity third: the dais floor and the ring shards overhead swap roles every count-in, the Regent's attacks continue from before, and the heart opens after two successful mechanic interactions per cycle. Lasers shorten the punish; a close kick is the big burst. No hidden bar, no unannounced instant kill. Defeat plays the return to Earth and the credits.

World structure



Stand-in Crownline City: roof decks, ivory towers, the elevated rail with its train, and the Thunderhead summit on the horizon.

A **region** is an open district about 2.4 km on a side with a main line of roughly 4 km through five chapters, entered at one edge and left at its landmark. It has three vertical bands: the **floor** (0–40 m: fields, streets, decks, basin sand), the **middle** (40–150 m: roofs, ledges, crane booms, arches) and the **sky** (150–350 m: tower tops, masts, rings, reefs), plus a boss or gate arena. The main line moves between bands every chapter, and the horizon landmark is placed so that it is visible from the floor of the first chapter.

Region	Entrance → exit	Peak climb	Landmark on the horizon
Sunseed Fields	0 → 25 m	60 m terraces	Crownline skyline
Crownline City	0 → 270 m	300 m observatory	Thunderhead summit
Thunderhead Range	0 → 190 m	110 m gorge, 300 m summit	The transmitter mast
Cinder Foundries	0 → –50 m	100 m casting trench	The crane forest
Tempest Docks	0 → 180 m	130 m upper gantry	The launch spine
Skyhook Works	0 → 300 m	–120 m shaft, 200 m ring	The star gate
Vermilion Basin	0 → 40 m	–90 m basin, 65 m terraces	The black sun
Cobalt Drift	0 → 330 m	330 m reef ascent	The drift moon

Region	Entrance → exit	Peak climb	Landmark on the horizon
Violet Inversion	0 → 160 m	160 m cathedral	The eclipse cathedral

Chapters

Each chapter is a leg between two landmarks with one dominant beat, keeping the 2D game's forty-five chapter names and their elevation envelopes converted to metres. A chapter is an interlude and the stronghold it walks toward, or the last stretch to the landmark: a checkpoint totem at the interlude's start and at the stronghold's edge, one signal on a high thing, one at the bottom of something and one in a cage, and a way back up from its floor. The last chapter of a region ends at the landmark with a stronghold, or with the mission boss.

Gravity

Zone	Gravity	Charged-leap apex	Use
Earth and industry	120 m/s ²	140 m	Baseline
Vermilion Basin	1.35×	104 m	Short, forceful arcs; wide ivory landings
Cobalt Drift	0.55×	255 m	Long glides between drifting reefs
Violet, normal	0.85×	165 m	Broad approach to the arches
Violet, inverted galleries	0.85× toward a ceiling	165 m	Optional ceiling lanes under arches; one required arch

Inversion is a volume, not a screen: passing a gravity gate or a seam under an arch flips local gravity for whatever is inside, Hopper rotates over 0.25 s to the new surface, the camera keeps the horizon, and the seam arrows show the way out. Cantors flip a marked volume beneath them; they never flip the whole district.

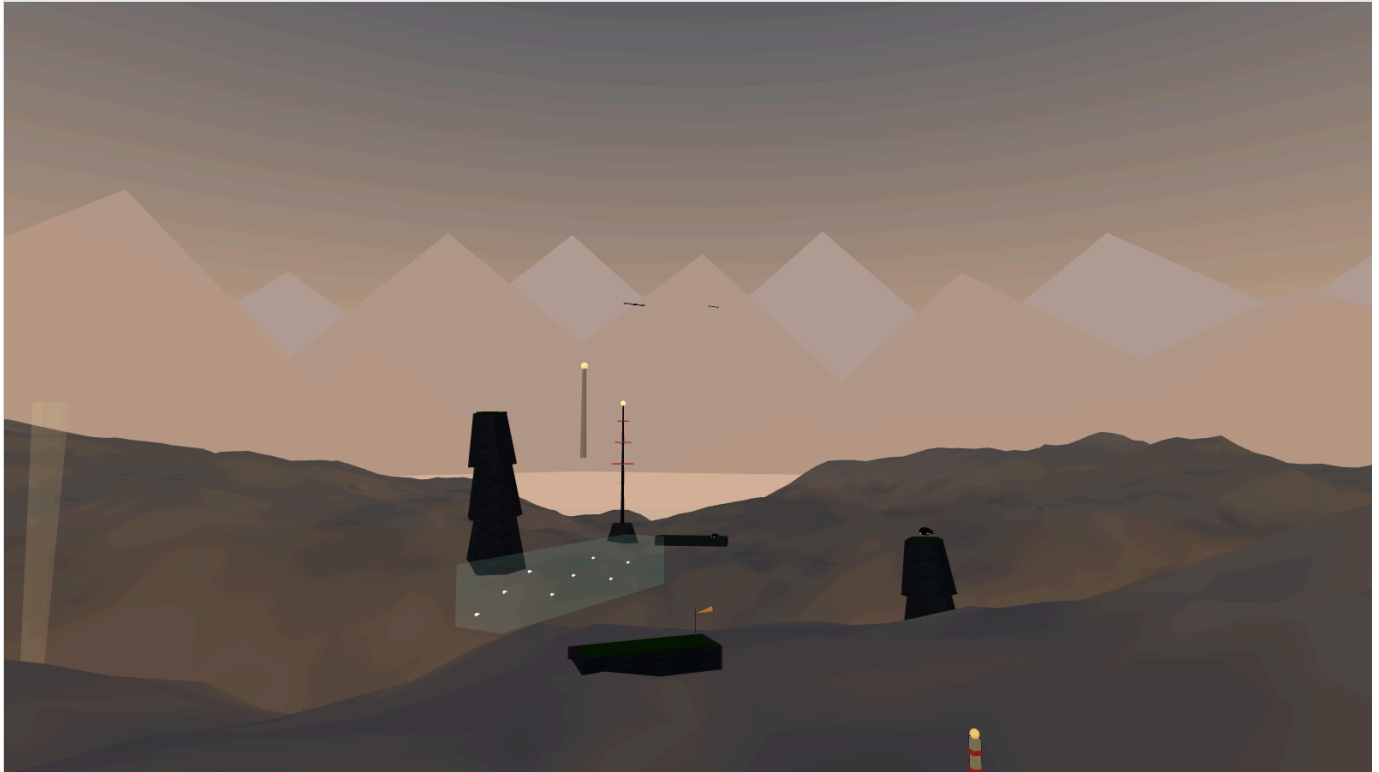
As built: a seam stands as a permanent flip volume from a hop's height up past its lintel, so walking under it does nothing and a hop enters it; inside, the controller steps Hopper in the world's mirror, so he falls up, lands on undersides, jumps down off them and climbs the piers back down. A turn halves his vertical speed, so a fall out of a seam's top settles at its edge. A jump away from a ceiling is a hop at seven tenths of the tap's kick, so it comes back to the lintel; the charged leap is the way to leave on purpose. Each seam's volume is hung under whichever lintel is really there, since a delivered arch is its own height rather than the height its stand-in was placed at. The commanders set the arena's gravity while they are awake; the Regent's third phase turns it upward toward the eclipse canopy over the dais. LEVEL-PLAN-M2-M3.md records the rest of what missions two and three built.

Checkpoints, signals, cages and capsules

Totems light when reached and are the respawn point. Nine signals per region unlock gallery entries and count on the results screen; three sit on high things, three at the bottom of things, three in cages that only

a reflected shot opens. Capsules sit where fights are. Spring pads are the region's elevators, thermals rise from chimneys, furnace doors and vents, and wind lanes cross open gaps.

Mission one · Earthbound Thunder



Stand-in Thunderhead Range: crag columns, a ravine bridge, a wind lane, a thermal and the transmitter on the summit.

Sunseed Fields

Honey-coloured morning, terraces, windbreaks, silos and fallen seed vessels. Three interludes and three strongholds: the irrigation lesson (a two-tier terrace on the road, a silo to wall-kick, a patrolling pair of hounds) walks toward the Orchard Granary, three silos close together with hounds crouched on their tops, rays dropping in and spitters rising by the path; the windbreak ridges (a thermal onto the ridge, the irrigation cut) walk toward the Seedfall Granary and its two vessels; the road to Crownline climbs a three-tier terrace toward the Crownline Gate. The Crownline towers stand on the horizon the whole time. Exit: the road to Crownline under the windbreaks.

Crownline City

Golden sunset, ivory towers with teal glass bands, roof decks, an elevated line with a train, a construction crown with a slewing crane, billboards that tip, and the observatory at the top. The main line climbs 270 m by roof decks, tower ledges and wall-kick corners, through three strongholds: the Ivory Ward (two towers with hounds on the roofs, rays dropping between them, leeches on the faces as the surge), the Construction Crown, and the Highline Observatory, with the roof ladder, the transit canyon (the rail span runs along the street, the train shuttling) and the sky bridges (a spring pad into a wind lane) as the interludes between. Exit: the observatory crown, looking at the summit.

Thunderhead Range

Slate, apricot storm cloud, an eclipse beginning over the summit. The slate descent steps down toward the Gorge Bastion, four crag columns rising out of the storm gorge past the rim with tortoises leaping from their tops and condors dropping over the edge; the broken ridge climbs out on a spring pad and a thermal toward the Cloudstep Bastion, a ring of crags around a low step; the summit road walks under the mast, visible from the gorge floor as a beacon, into the Night Rook's arena. Tortoises hold the shelves; condors make the ascent an air fight.

The mission's kit

Terrace steps, farmhouses, silos, windbreaks and seed vessels; ivory towers, roof decks, rail spans, train cars, the construction crown, billboards and the observatory; crag columns, ledge shelves, ravine bridges, windsocks and the transmitter. Seventeen structures, requests M-024 to M-040.

Mission two · The Iron Migration



Stand-in Tempest Docks: breakwater, container stacks, crane booms, a gantry tower, the freighter and the launch spine beyond.

Cinder Foundries

Ochre smoke and furnace light. Furnace towers with glowing doors that are thermals, conveyor spans that add their velocity to Hopper, chimneys with a thermal above each lip, slag barges on a casting channel whose hot slag lifts anyone who falls in, and stamping presses whose rams fling rather than crush. Furnace hounds charge along the belts; casters lob slag that cools into stepping stones over the channel. The route goes down through the casting trench and out toward the crane forest.

Tempest Docks

Rain, sea-green steel, white spray. Container stacks are stairs; crane booms slew and carry containers on cables; the gantry tower has four decks and a bridge; the freighter on the storm channel is the chapter's long crossing; the sea beside the breakwater pushes a fallen Hopper ashore. Mantas drag the platforms Hopper is about to land on until their tether nodes are shot; crabs slam the decks. The launch spine stands over the channel and the route climbs the crane forest toward it.

Skyhook Works

Pale gold above the clouds. The exhaust shaft descends 120 m by baffles with a thermal at the bottom; piston stairs rise and fall out of phase; the gantry elevator rides 140 m beside the rocket; the launch ring crowns the spine 200 m up. Coil wraiths gate lanes between scaffolds until a node is shot; turbine wasps

dash and, kicked, stall into platforms. The Leviathan is coiled around the elevator tower, and the star gate opens when it dies.

The mission's kit

Furnace towers, conveyor spans, chimneys, slag barges, stamping presses; crane booms, container stacks, the freighter, gantry towers, breakwaters; launch rings, rockets, piston stairs, exhaust shafts, gantry elevators. Fifteen structures, requests M-041 to M-055.

Mission three · Beyond the Black Sun



Stand-in Cobalt Drift: floating reefs on drifting paths, root pillars, a dust current and the moon.

Vermilion Basin

Red coral, ivory ribs, a black sun. Gravity is 1.35×, so leaps are short and landings heavy; the ivory-topped terraces are wide on purpose. Rib arches are bridges and choir perches; coral spires are cover; the staged coral bridge breaks behind Hopper in three signalled stages, dropping him to the basin floor if he dawdles, where a spring pad waits. Burrowers tunnel under the landings; choirs sing fans from the arches. The route descends into the basin and climbs the terraces toward the eclipse.

Cobalt Drift

Ultramarine void, stars, a huge moon. Gravity is 0.55× and everything floats: reefs drift on marked dust-current paths, root pillars rise 160 m with shelves, and a 330 m ascent is made by glides, currents and medusa bounces. Medusae are the springiest stomps in the game; skates blink and dash. Falling lands in blue dust that lifts Hopper gently back to the lowest reef. The moon is the landmark.

Violet Inversion

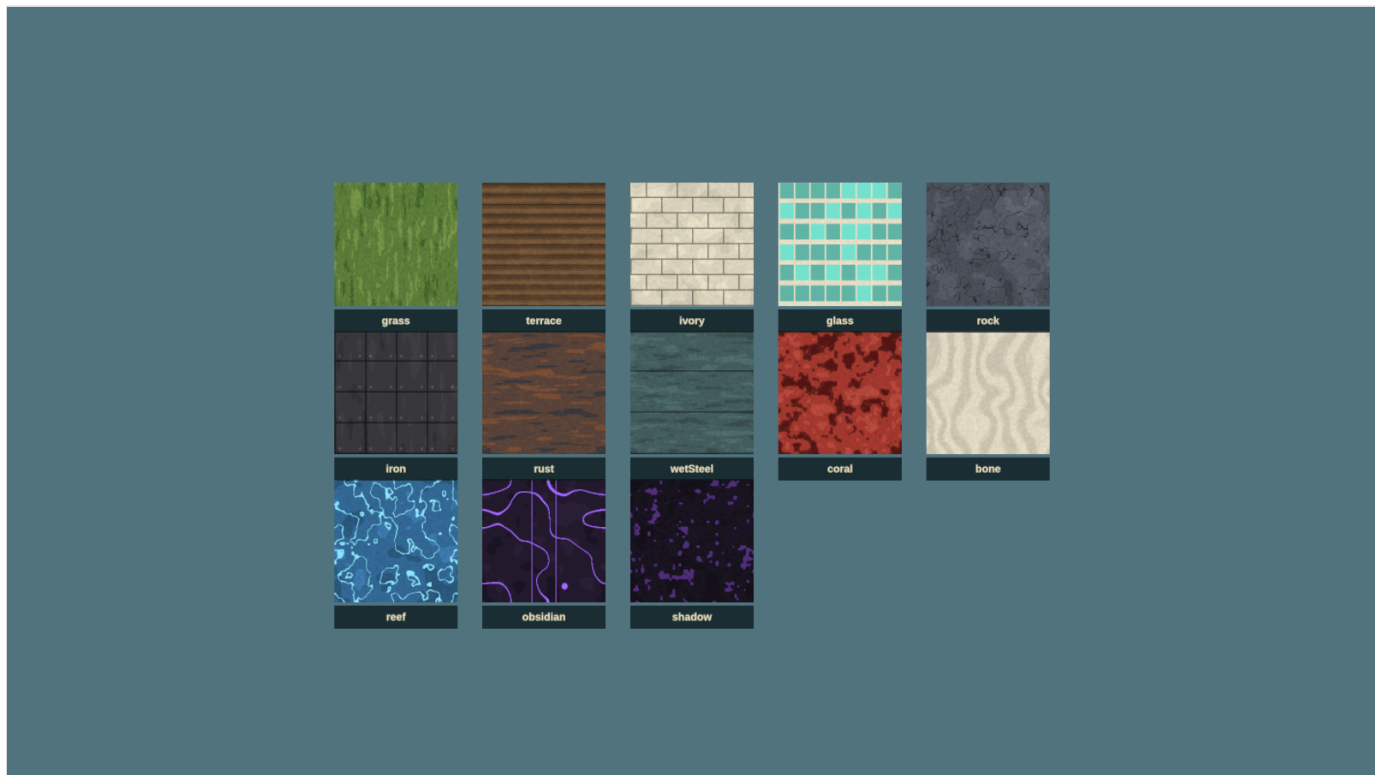
Dark purple, obsidian arches, ring shards orbiting slowly, glowing gravity seams. Gravity is 0.85×; under every arch a seam marks an optional inverted lane on the lintel's underside, and one arch late in the region must be crossed inverted. Ring shards are landings that rotate through the sky. Stalkers walk

floors and ceilings; cantors flip volumes. The cathedral is visible from the first arch and reached at the eclipse dais, where the Regent descends.

The mission's kit

Ivory rib arches, coral spires, ivory-topped terraces, staged coral bridges; floating reefs, root pillars, dust currents; obsidian arches, ring shards, the cathedral facade, gravity seams, the eclipse dais. Twelve structures, requests M-056 to M-067.

Visual direction



The procedural stand-in textures: flat tones, brushed grain, no photographic detail.

The 2D game's look is 1970s cel animation for characters over gouache landscapes. In 3D that becomes:

- **Cel shading** with a three-tone ramp on characters and structures and a four-tone ramp on terrain, and **ink outlines** by inverted hull on characters and props. No specular, no normal-map micro detail, no physically based gloss.
- **Painted skies**: one equirectangular painting per region with the sun or eclipse placed where the light comes from, seen from inside a dome. The sky is most of every screenshot and carries the region's mood.
- **Gouache textures**: hand-painted tileable albedo with visible brush direction and two or three flat tones. Terrain is vertex-painted in bands by height and slope, so the painting and the collision are the same surface.
- **Haze bands**: exponential fog in the region's haze colour, and two rings of painted horizon cards at 5 km and 6.75 km in the haze tones, so distance reads as it does in the concept boards.
- **Shadows** are only cast by Hopper and by shadows; the world is lit by the painted sky, a directional sun and a hemisphere fill in the region's own colours.

Palette

The nine region palettes are the exact hex values the 2D game uses, so a 3D region reads as the same place as its painted background.

Region	Sky	Haze	Accent	Ground
Sunseed Fields	#739cae	#f8c479	#ffdf7a	#456747
Crownline City	#b98687	#f6bd86	#a1efe3	#465663
Thunderhead Range	#7c8d9c	#d5ad8d	#e9d09b	#535963
Cinder Foundries	#644d45	#c28754	#ffb454	#4d4345
Tempest Docks	#667d84	#b1c0b5	#b4f2df	#3e5d61
Skyhook Works	#a5a9a5	#efe0bd	#ffd38c	#716566
Vermilion Basin	#6c2533	#de765d	#fff0bc	#9a4348
Cobalt Drift	#193d77	#478bab	#b9fff1	#355d91
Violet Inversion	#251633	#60436c	#e5b8ff	#503653

Shadows keep their language: charcoal interiors, violet contours, pale cores; furnace species glow orange. Hopper keeps his green, cream, red, blue and gold. Saturated red-white is reserved for lasers, amber-white for danger tells, teal for the kick's arc, violet for gravity.

Scale cues

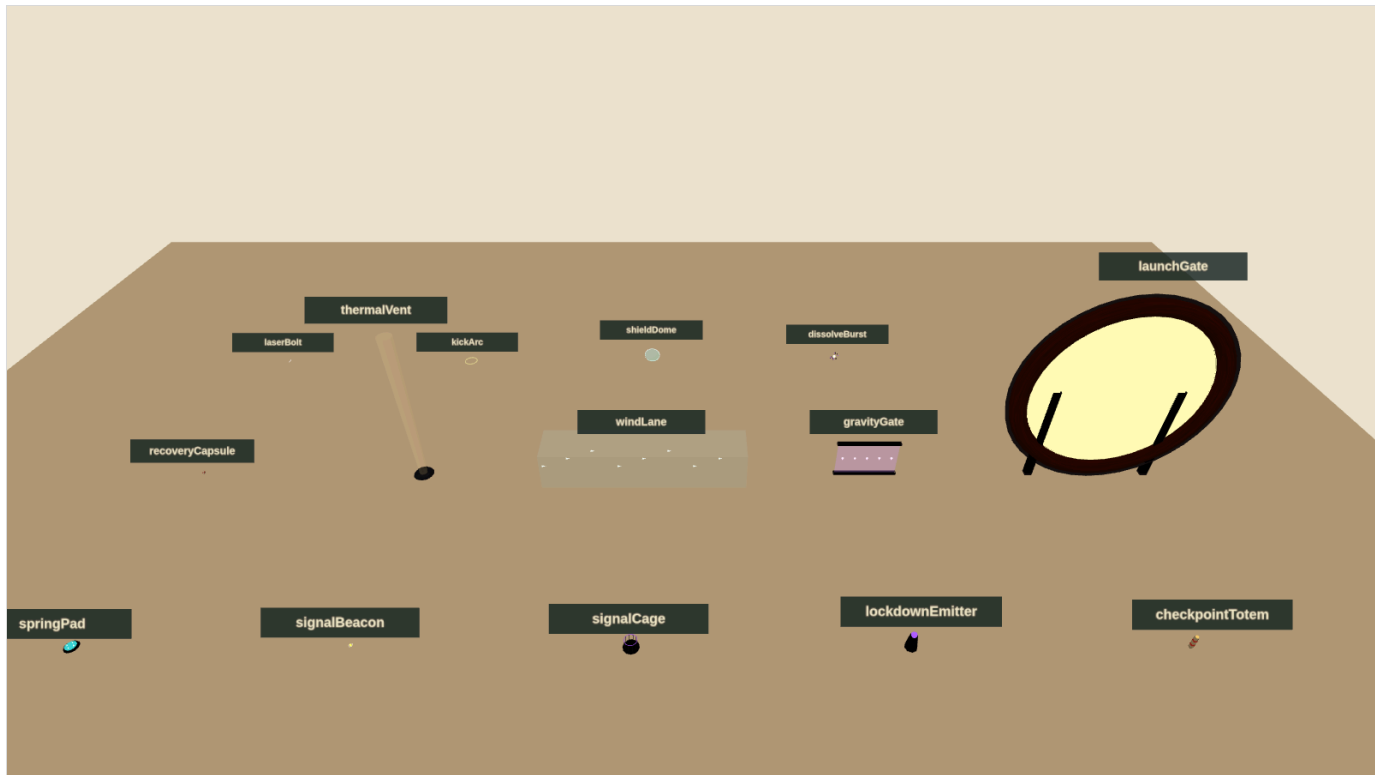
Hopper is huge, and the world must keep saying so: farmhouses to his knees, train cars he can stand on, containers as stairs, windbreak trees at his shoulder. Every kit piece is built to a real size and the stand-in sizes in the request documents are the model sizes.

Structures, props and the stand-in policy



Stand-in structure kit for mission one on a contact sheet; industry and alien kits have their own sheets in design/assets.

Forty-four structure pieces in nine kits, one trim sheet per kit, are what the districts are built from. Each piece publishes its **landings**, the flat tops a route may use, and the level tools place jumps against those, not against the mesh, so a stand-in and its final model are interchangeable if their landings agree within half a metre. Moving pieces (train, crane boom, piston heads, elevator cage, press ram, barge, reefs, ring shards) are rigid hierarchies driven by transforms.



Stand-in props: spring pad, signal, cage, lockdown emitter, checkpoint totem, capsule, thermal, wind lane, gravity gate, launch gate and the effects.

Stand-ins

Every model, texture and sky in the request documents has a procedural placeholder, built from primitives with cel materials and ink hulls, in `hopper/3d/standins/`. They are the same size as the requested model, expose the same named sockets, publish the same landings and animate a little, so gameplay code binds to names that survive the swap. Textures are painted in code into data textures, skies are gradient-plus-sun paintings, terrain is a heightfield whose colour and collision share one function, and the horizon is a ring of silhouettes in the haze colour.

The creature stand-ins are shaped to the turnaround sheets in `design/references` rather than left as blocks: plated bodies drawn in profile and extruded, rib arcs, crests, claws and horns cut from outlines, tapered strands for tails and tendrils, lathes for bells, pots and shells, feathered wings on jointed bones. They are still shadow — black plates, violet seams, ivory teeth and cores — so the game can play with them until the delivered models arrive.

The manifest maps each request to its stand-in and its final file. When an asset arrives, its status flips from `stand-in` to `delivered`, the loader takes the GLB or texture, and the stand-in stays available for tests. A build with any stand-ins left shows a small "stand-in art" mark in the corner and must not be the public build.

Title screen, menus, HUD and sound

The title screen is the one the 2D game ships: the painted poster with the boy on Hopper's back, the generated logo, Press Start, the play-select screen with Continue and the three episodes, and Instructions, Settings and Fullscreen at the bottom right. The 3D edition uses the same React components and the same poster; the only visible difference is the controller diagram, which shows the 3D mapping, and a "3D" mark on the episode list. Fullscreen on start, any fresh controller button from the idle title, left stick for utility navigation, all as the collection guidelines require.

HUD

The top middle of the frame is the way ahead and stays empty: nothing informational is ever drawn there. The top corners hold what they did in 2D, six armour pips with the laser heat gauge beneath them at the top left, the signal count and the sound, fullscreen and pause buttons at the top right. Everything else sits low. The episode, region, chapter, route progress and the landmark compass line are at the lower left above the button legend; the stronghold host count, the locked target's name and health and any gravity note are at the lower right; the commander's bar, phase and tell sit along the bottom edge in the middle; the district title card rises in the lower third. In the world itself: the landing guide ring, height ticks at the right-hand screen edge while airborne, the crosshair in the middle of the aimed picture and the kick's arc sweeping around Hopper while the spin runs. Boss health appears only in boss arenas. Nothing sits where the rider, the next landing or the road ahead would be.

Pause and settings

Pause keeps Resume, Instructions, Settings, Restart Checkpoint and Return to Title with the confirmation rules. Settings add camera sensitivity and inversion for both axes, field of view, landing guide (always, airborne only, off), lock-on style (hold, toggle), reduced motion, vibration, and the existing master, music and effects levels, dead zones, subtitles and high-contrast threat outlines. Preferences persist locally.

Music and sound

The three supplied recordings keep their roles: **Hopper the Grasshopper** on the title and menus, **Grass March 1** through missions one and two, **Dark Moon** through mission three. Track positions are preserved across scenes, tabs and pauses, and pause ducks rather than switches, exactly as the shipped audio module does; the 3D edition reuses that module. Effects gain position and distance: laser pulses, kicks, stomps, dissolves and enemy tells are placed in the world so an ambush from behind is heard behind. Wind rises with height and speed in a glide. Boss telegraphs duck the music briefly. Every mandatory cue has a visual equivalent.

Technical plan

As built: every delivered structure carries collision baked from its own GLB

(hopper/3d/models/collision.json, made by models/source/bake-collision.mjs: a 2 m grid of tops and undersides plus a ring scan for near-vertical faces, merged into boxes in the model's frame), and the world uses those boxes instead of the stand-in's primitives wherever a delivered file exists. The audit that motivated it (models/collision-audit.md) found stand-in and delivered surfaces disagreeing by up to 120 m, which was the falling-through and the invisible floors. Signals authored on a designed landing that the real model no longer has are moved onto the nearest real top, totems and capsules are lifted out of a body that now stands where they were, and any of them under an overhang lower than Hopper slides sideways to open air.

The 3D edition is a second entry in the same repository, sharing the shell, input and audio code with the 2D game and adding a 3D world, simulation and renderer.

- **Rendering:** three.js (r185, the version the stand-ins use) over WebGL2. MeshToonMaterial with ramp textures, inverted-hull outlines, exponential fog, one shadow-casting light for Hopper and nearby shadows, sRGB output. Instanced kit pieces; three LOD levels per structure; painted horizon cards and sky dome; streaming in 400 m cells around the camera.
- **Models:** the delivered GLBs loaded with the meshopt decoder and KHR_mesh_quantization, as models/README.md describes; final assets follow the same conventions. The rider binding helper in models/attachments.js is used for the separate files; the combined file needs no binding.
- **Simulation:** fixed 120 Hz steps as in the 2D engine. Hopper is a capsule (5 m radius, 14 m tall) against a heightfield sampled from the region's heightmap plus box and convex colliders from the structure kits. Fast states (dive, super leap) sub-step four times so nothing tunnels at 90 m/s. Moving platforms carry what stands on them.
- **Enemies:** state machines with tells, attacks and recoveries in the manner of the shipped combat module, on a **perch graph** of landings rather than a navmesh: ground shadows move between landings they can reach, flyers along authored lanes. Waves and knots as described; a stronghold's host waits on the perch graph's tops in view.
- **Gaits:** a small pure module decides which of the three gaits is running, keeps the stride clock, and puts the six feet in the world; a rig module solves two-bone IK up each leg on the delivered skeleton to reach them. Both run in the node tests at the game's own rate, so the rhythms, the suspension in the gallop, the planting and the body's angle on a slope are checked rather than eyeballed.
- **Camera:** a follow rig with spring collision, the aiming and Horizon View rigs, the climb framing, and the leap and glide framing rules above.
- **Input and audio:** the shipped input.ts (all connected pads combined, keyboard fallback, connection handling) with the right stick, triggers and bumpers added; the shipped audio.ts with a panner per effect voice.
- **Shell:** the shipped React title, play-select, HUD, pause, settings and instructions with the 3D diagram and the new settings.
- **Saves:** local, per checkpoint, with cleared knots, signals and settings.

- **Build:** a second static entry, `hopper/3d/index.html`, built by the same Vite Pages configuration and published at `/mini/hopper/3d/` beside the 2D game. The stand-in viewer is already published at `/mini/hopper/3d/viewer/`.

Performance budget

Item	Budget
Target	60 fps at 1080p on an integrated GPU; 30 fps floor on a 2019 laptop
Draw calls	under 600 in a district view; instancing for kit pieces
Triangles on screen	under 1.5 M, with LOD 1 beyond 300 m and cards beyond 1.2 km
Textures resident	under 256 MB per region; skies as KTX2
Simulation	under 2 ms per frame including enemies
Load	a region streams in under 4 s from a warm cache

Testing

- **Simulation tests in node**, like the shipped engine tests: the jump family reaches its stated apexes and ranges in every gravity; falling never damages; the way-back-up rule holds from every floor sample; enemies' tells last their stated time.
- **Route audits**: for every chapter, every required landing pair is within the jump model with a 25% margin; every knot has a visible route around it; every floor has a return within 20 s at run speed.
- **Stand-in tests** (already in place): every request's stand-in builds, exposes its sockets and landings, matches its size, and the manifest and registry agree.
- **Visual QA** with headless renders (the figures in this document are made that way) and browser checks on the agreed matrix, plus physical Xbox testing recorded separately from simulated input.

Asset plan

The asset requests are separate documents so they can be worked and updated independently: `model-requests.md` (93 models) and `image-requests.md` (47 images and textures, round one) and `image-requests-round-2.md` (32 model reference sheets, round two), both generated from `source/build_requests.py` together with `standin-manifest.json`.

Family	Requests	State
Hopper and rider	1 delivered model; 2 clip requests	Models delivered; new clips open
Shadow species	18 skeletal, blend-shape, spline or rigid rigs	Stand-ins in place
Commanders	3 bespoke rigs	Stand-ins in place
Structure kits	44 pieces in 9 kits, static or rigid	Stand-ins in place
Props and effects	15	Stand-ins in place
Landmarks and terrain	9 landmarks; 9 terrain sculpts	Stand-ins in place
Skies, horizons, terrain sets, trim sheets	9 of each	Procedural placeholders in place
Reference sheets (turnarounds, kit sheets, props, Hopper poses)	32	Open; generated before the models they describe
Interface and effects images	11	3 delivered (poster, cards, 3D diagram); ramps procedural; 7 open

Rigging in one page

Every model request names one of five rig types. **Skeletal** rigs (hounds, tortoises, crabs, casters, stalkers, the Rook and the Regent) are skinned skeletons with listed joints, clips and sockets. **Blend shapes** on skeletal rigs give the organic tells: a spitter's throat, a choir's heads, a medusa's bell. **Spline chains** (leech, wraith, the Leviathan) are bone chains the game drives along curves. **Rigid hierarchies** (wasp, cantor, skate, cranes, pistons, gates) are parts on named pivots moved by transforms. **Static** meshes (most structures) have no rig; belts scroll their UVs. Every creature has `Core`, `Hitbox`, `Body` and a `Mouth` or `Emitter`, and everything stompable has a `Landing` socket. The Hopper rig already exists; its requests are clips only.

References before models

No model is started from a single side-view sprite. Every model request names the art it must match and a reference sheet that is generated and approved first: a turnaround for each species and commander, a kit sheet per region drawn at a shared scale with Hopper, a props sheet, and a pose sheet for Hopper's

new clips. These are the second round of image requests, `image-requests-round-2.md`, added after the first round had begun; it maps every sheet to the models that wait on it.

Order of arrival

Art can arrive in any order because the stand-ins hold every slot, but the order that helps most is: the reference sheets for mission one, the Hopper clips (glide and dive change how the game feels), the four mission-one species, the Sunseed and Crownline kits and skies, the Night Rook, then the rest by mission. Paintings should be checked in the viewer against the stand-in they replace before they are committed.

Production plan

Milestone	Deliverable	Exit criteria
M0 · Design and stand-ins	This document, the request documents, the stand-in library and viewer	Delivered with this document
M1 · First leap	Sunseed district playable: controller, camera, the jump family, glide, dive, wall kick, mantle, landing guide, terrain and kit collision, the animated GLB	Ten minutes of free roaming feels good to three people; the skyline is visible from the start; nothing kills
M2 · Shadows	Combat core with hound, spitter, ray and leech; armour, lasers, kick, parry, stomps, dive, lock-on, knots, dissolves	Air-stomp chains work; a knot fight is fun on standard and readable on relaxed
M3 · Earthbound Thunder	Three districts authored with the kit, chapters, totems, signals, cages, capsules, the Night Rook; title, play-select, HUD, pause, settings, save	Mission one playable start to finish; route audit and simulation tests pass
M4 · The Iron Migration	Moving structures, belts, thermals, wind, tethers, presses; six species; the Leviathan	Mission two playable; performance budget met in the docks
M5 · Beyond the Black Sun	Gravity volumes, drifting reefs, inversion galleries, six species, the Regent, transitions, ending	Full campaign playable; Dark Moon routed
M6 · Polish	Streaming, LOD, accessibility, instructions, controller QA on the browser matrix, results and gallery	Acceptance checklist of the collection guidelines passes
M7 · Art swap	Final models, skies and textures replacing stand-ins as they arrive, from M1 onward	No stand-in mark in the public build

Milestones one and two are the ones to spend tuning time on; everything after them is content on a proven feel. Mission one can ship alone if the campaign needs to be staged.

Risks

Risk	Mitigation
Landing in 3D feels uncertain	Landing guide, flat-top language and height ticks from M1; measured in the M1 test
Camera inside dense structures	Spring collision, landings kept 10 m from walls, kits designed with open corners
Performance on integrated GPUs	Toon shading is cheap; instancing, LOD and cards from M3; budget checked each milestone
Ninety-three models	Kits share trim sheets; stand-ins hold every slot; mission one first
Open districts lose the route	The landmark rule, Horizon View, the compass strip, totems at every chapter

Risk	Mitigation
Only-shadows-hurt makes hazards toothless	Hazards move Hopper a long way; knockback off a roof is the real cost

Decisions for this review

- The four new verbs and their buttons: glide on held A, dive stomp on Y, crouch charge on RB, Horizon View on LB, lock-on on LT.
- District size (2.4 km) and the ten-minute region.
- Falling never hurts; hazards push and never damage; knockback is the cost.
- An all-round kick that is the whole defence: it parries from any side, and there is no held guard.
- Shipping as a second entry beside the 2D game with the same title screen.
- The order in which art should arrive.

Delivered with this document

The design text and its Word and PDF renders; `model-requests.md` with rig types for all 93 models; `image-requests.md` for 47 paintings and textures; `standin-manifest.json`; the stand-in library with tests; the review viewer with nine dioramas and seven contact sheets, published beside the game; the 3D controller diagram; and the rendered figures in `design/assets/`.